

Supplement: From Accuracy to Allocation

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1 Extended Notes on Data Provenance

Sources and licenses.

- **New York:** *311 Service Requests from 2010 to Present*, NYC Open Data, dataset `erm2-nwe9`; NYC Open Data terms (free use without restriction).
- **Chicago:** *311 Service Requests*, Chicago Data Portal, dataset `v6vf-nfxy`; covers the current 311 system live since 2018-12-18 (the study window starts 2019-01-01 to avoid the system transition); source-flagged duplicates dropped.
- **San Francisco:** *311 Cases*, DataSF, dataset `vw6y-z8j6`; Public Domain Dedication and License.
- **Austin:** *Austin 311 Public Data*, City of Austin Open Data Portal, dataset `i26j-ai4z`.
- **Weather:** NOAA GHCN-Daily through the AWS Open Data Program mirror `noaa-ghcn-pds` (maintained by NOAA NCEI); stations USW00094728 (Central Park), USW00094846 (O'Hare), USW00023272 (San Francisco Downtown), USW00013904 (Camp Mabry). Native units (tenths of mm/°C, tenths of m/s) converted only in preprocessing.

Acquisition queries. Each 311 source is queried through the SODA API with server-side aggregation — `$select=date_trunc_ymd(created_date), category, count(*)`, grouped and ordered, chunked by calendar year, paginated at 50,000 rows — so the rawest stored layer is already free of record-level attributes. The complete list of issued query URLs for every city is in `data/raw/311/*manifest.json`.

Known data characteristics. Reported request volumes reflect intake-channel structure (phone/app/web), administrative practice, and reporting propensity, all of which differ across cities and over time; the Chicago information-only and aircraft-noise exclusions and the SF duplicate exclusion are documented with reasons in `configs/service_families.yml`. A missing (day, family) pair within a city’s observed range denotes zero requests created, and the panel densification step materializes those zeros explicitly.

2 Service-Family Harmonization

The harmonization rule set (`configs/service_families.yml`) is an ordered, case-insensitive substring matcher: the first matching family wins, and categories matching no rule fall to *other*. Order resolves genuine ambiguities deterministically (e.g., “noise” complaints about construction are noise, not housing/building, because the noise rule precedes). The full mapping from every native category to its family, with request counts, is exported per city to `outputs/tables/family_composition_{city}.csv` by the preprocessing stage and ships with the repository; the tables below summarize the dominant native categories per family in each city.

Two properties matter for interpretation. First, the rules are *versioned*: any change to the mapping is a visible diff, and the unit test suite pins representative mappings. Second, family compositions are *not* identical across cities — e.g., the noise family is dominated by residential-noise complaints in New York but is a much smaller share elsewhere; the sanitation family includes encampment-related categories in San Francisco that have no Chicago counterpart. All cross-city statements in the main text are therefore about families as *planning buckets within each city*, and the cross-city axis compares the structure of the accuracy-to-decision mapping, never raw family volumes.

Austin

- **housing building** (1.9%): DSD - Outdoor Commercial Venue Music Complaint (7,867); Park Maintenance - Grounds Plumbing Issues (3,529); ATD - Construction Items in Right of Way (3,132)
- **noise** (4.7%): APD - Non Emergency Noise/Alarm (70,197)
- **other** (23.5%): Loose Dog (27,062); 311 CC - Other (25,843); Animal Protection - Loose Dog (24,018)
- **parks trees** (4.4%): Park Maintenance - Grounds (37,444); Tree Issue ROW (13,396); PWD - Tree Issue Right of Way (5,068)
- **public safety** (3.2%): APD - Non Emergency Collision (32,924); Sign - Traffic Sign Emergency (2,185); ZZ - AFD - Fireworks Complaint (2,049)
- **sanitation** (21.3%): ARR - Garbage (79,958); ARR - Compost (49,879); ARR - Storm Debris Collection (49,751)

- **streets transport** (38.5%): Austin Code - Request Code Officer (81,975); DSD - Request Code Officer (47,058); ATD - Parking Violation Enforcement (42,161)
- **water sewer** (2.6%): AW - Water Conservation Violation (10,628); WPD - Channels/Creek/Drainage Issues (7,643); AW - Water Waste Report (3,978)

Chicago

- **housing building** (6.9%): Building Violation (107,110); Water Lead Test Kit Request (65,635); Buildings - Plumbing Violation (40,969)
- **other** (6.9%): Stray Animal Complaint (61,065); Report an Injured Animal (50,921); Vicious Animal Complaint (37,396)
- **parks trees** (10.3%): Tree Trim Request (NO LONGER BEING ACCEPTED) (206,525); Weed Removal Request (108,205); Tree Removal Inspection (96,345)
- **public safety** (2.6%): Tree Emergency (121,023); Commercial Fire Safety Inspection Request (1,557)
- **sanitation** (38.3%): Graffiti Removal Request (556,086); Rodent Baiting/Rat Complaint (295,754); Garbage Cart Maintenance (282,964)
- **streets transport** (28.7%): Abandoned Vehicle Complaint (269,189); Pothole in Street Complaint (215,010); Street Light Out Complaint (197,946)
- **water sewer** (6.3%): Water in Basement Complaint (56,251); Sewer Cleaning Inspection Request (52,303); Check for Leak (45,813)

New York

- **housing building** (20.5%): HEAT/HOT WATER (1,439,229); UNSANITARY CONDITION (605,801); PLUMBING (343,419)
- **noise** (23.3%): Noise - Residential (2,287,530); Noise - Street/Sidewalk (1,027,860); Noise - Vehicle (374,275)
- **other** (6.2%): GENERAL (166,982); Consumer Complaint (138,005); Maintenance or Facility (121,806)
- **parks trees** (2.9%): Damaged Tree (214,620); Overgrown Tree/Branches (115,202); New Tree Request (93,700)
- **public safety** (2.7%): Non-Emergency Police Matter (190,623); Illegal Fireworks (112,922); Drug Activity (86,992)
- **sanitation** (11.5%): Request Large Bulky Item Collection (632,148); Dirty Condition (224,713); Rodent (220,299)
- **streets transport** (28.5%): Illegal Parking (2,478,000); Blocked Driveway (934,597); Street Condition (410,510)
- **water sewer** (4.4%): Water System (366,096); WATER LEAK (220,757); Sewer (172,464)

San Francisco

- **housing building** (0.6%): Residential Building Request (11,712); General Request - BUILDING INSPECTION (7,468); Residential Building (6,364)
- **noise** (1.3%): Noise Report (37,785); Noise (18,133)
- **other** (8.8%): General Request - PUBLIC WORKS (105,723); General Request (62,711); General Request - MTA (54,573)
- **parks trees** (2.8%): Tree Maintenance (80,947); Rec and Park Requests (40,539)
- **public safety** (0.0%): General Request - FIRE DEPARTMENT (490)
- **sanitation** (19.6%): Graffiti (337,617); Encampments (158,952); Graffiti Public (121,575)
- **streets transport** (65.4%): Street and Sidewalk Cleaning (1,933,456); Parking Enforcement (602,068); Abandoned Vehicle (62,042)
- **water sewer** (1.6%): Sewer Issues (47,683); Sewer (18,406); Catch Basin Maintenance (2,161)

3 Model Configurations

Hyperparameters are fixed across cities and feature sets so that no configuration receives a larger tuning budget than another; values were chosen a priori in the ranges standard for daily-count panels of this size and are not per-task optimized.

Model	Parameters	Notes
Trailing mean	7-day window	forecast-free baseline
Seasonal naive	same weekday, $t - 6$	forecast-free baseline
Ridge	$\alpha = 1.0$, standardized inputs	
Random forest	300 trees, min leaf 5, 60% features	Breiman (2001)
LightGBM point	L1 objective, 600 trees, lr 0.05, 63 leaves, min child 30	Ke et al. (2017)
LightGBM quantile	as above, pinball objective per level non-crossing via cumulative max	levels .05/.25/.50/.75/.95

Categorical handling: one-hot family indicators everywhere; one-hot city indicators in pooled (global) models only. Predictions are clipped at zero. The selected configuration per (scope, city, feature set, model) is refit on train+validation before its single test evaluation, exactly as in the protocol of the main text.

4 Allocation Optimality

The greedy allocation used by all expected-value policies is exactly optimal for the per-day problem it solves.

Proposition 1. *Let $D_1, \dots, D_S$ be non-negative random variables, $\omega_s > 0$, $\kappa > 0$, and consider*

$$\max_{x \in \mathbb{Z}_{\geq 0}^S, \sum_s x_s = B} \sum_{s=1}^S \omega_s \mathbb{E}[\min(D_s, \kappa x_s)].$$

The greedy algorithm that assigns crews one at a time to the family with the largest marginal gain attains the optimum.

Proof. For fixed s , define $g_s(x) = \omega_s \mathbb{E}[\min(D_s, \kappa x)]$ on integers $x \geq 0$. For any realization $d \geq 0$ the map $c \mapsto \min(d, c)$ is concave in c ; expectation preserves concavity, and restriction to the integer grid yields non-increasing increments $\Delta_s(x) = g_s(x+1) - g_s(x)$. The problem is therefore the maximization of a separable discrete-concave function over the simplex $\sum_s x_s = B$, for which the greedy/incremental algorithm is optimal: at every step the chosen increment is at least as large as any increment available later (by monotonicity of Δ_s), so an exchange argument converts any optimal solution into the greedy one without loss of objective value. $\square$

In the implementation $\mathbb{E}[\min(D_s, c)]$ is evaluated exactly for the discrete distribution defined by linear interpolation of the inverse CDF through the five forecast quantiles on a fixed 99-point grid (`src/optimization/allocation.py`); point forecasts use a degenerate distribution, and the oracle uses the realized value. The unit test `test_greedy_matches_bruteforce_on_small_instance` verifies greedy-vs-brute-force equality on random instances.

5 Full Result Tables

All tables regenerate from `scripts/make_tables_figures.py`; CSV versions live in `outputs/tables/` with provenance hashes in `_provenance.json`.

Validation-selected headline (test MAE of the validation-chosen model per city and feature set):

city	internal	calendar	weather	calendar_weather
austin	16.529	14.843	16.530	14.844
chicago	60.374	50.945	58.665	49.295
nyc	207.022	182.378	186.728	165.417
sf	35.183	31.532	35.866	30.061

Selection detail (chosen model, validation MAE, single test evaluation):

city	feature_set	selected_model	val_mae	test_mae
austin	calendar	lgbm_point	15.389	14.843
austin	calendar_weather	lgbm_point	15.245	14.844
austin	calendar_weather_lagged_only	lgbm_point	15.711	14.865
austin	internal	random_forest	16.494	16.529
austin	weather	lgbm_point	16.462	16.530
chicago	calendar	lgbm_point	48.527	50.945
chicago	calendar_weather	lgbm_point	46.802	49.295
chicago	calendar_weather_lagged_only	lgbm_point	47.997	50.388
chicago	internal	random_forest	54.803	60.374
chicago	weather	lgbm_point	52.580	58.665
nyc	calendar	lgbm_point	146.497	182.378
nyc	calendar_weather	lgbm_point	131.660	165.417
nyc	calendar_weather_lagged_only	lgbm_point	150.323	178.626
nyc	internal	lgbm_point	165.701	207.022
nyc	weather	lgbm_point	153.963	186.728
sf	calendar	lgbm_point	30.341	31.532
sf	calendar_weather	lgbm_point	28.973	30.061
sf	calendar_weather_lagged_only	lgbm_point	30.173	31.659
sf	internal	ridge	31.435	35.183
sf	weather	ridge	31.850	35.866

Exhaustive descriptive grid (every model $\times$ feature set; never used for selection):

city	model	internal	calendar	weather	calendar_weather
austin	lgbm_point	16.719	14.843	16.530	14.844
austin	naive_trailing7	30.371	30.371	30.371	30.371
austin	random_forest	16.529	15.506	16.617	15.467
austin	ridge	24.976	25.677	24.955	25.609
austin	seasonal_naive7	21.613	21.613	21.613	21.613
chicago	lgbm_point	59.236	50.945	58.665	49.295
chicago	naive_trailing7	111.461	111.461	111.461	111.461
chicago	random_forest	60.374	54.418	59.268	53.741
chicago	ridge	71.809	70.510	72.851	70.810
chicago	seasonal_naive7	84.238	84.238	84.238	84.238
nyc	lgbm_point	207.022	182.378	186.728	165.417
nyc	naive_trailing7	305.618	305.618	305.618	305.618
nyc	random_forest	209.281	195.088	200.152	190.629
nyc	ridge	215.154	213.486	214.625	213.113
nyc	seasonal_naive7	282.439	282.439	282.439	282.439
sf	lgbm_point	35.046	31.532	33.225	30.061
sf	naive_trailing7	44.769	44.769	44.769	44.769
sf	random_forest	34.515	32.840	33.215	31.562
sf	ridge	35.183	36.562	35.866	36.691
sf	seasonal_naive7	43.786	43.786	43.786	43.786

Quantile-forecast evaluation:

scope	city	model	pinball	coverage_90	coverage_50	mae_median
local	austin	lgbm_quantile	5.162	0.812	0.413	14.790
local	chicago	lgbm_quantile	17.285	0.768	0.375	48.959
local	nyc	lgbm_quantile	58.980	0.769	0.342	167.194
local	sf	lgbm_quantile	10.189	0.808	0.344	29.286
global	austin	lgbm_quantile	4.991	0.872	0.485	15.169
global	chicago	lgbm_quantile	17.618	0.831	0.456	52.561
global	nyc	lgbm_quantile	58.862	0.818	0.408	170.492
global	sf	lgbm_quantile	9.748	0.745	0.399	29.611

Rank agreement between accuracy and simulated-decision orderings (descriptive):

city	regime	n_models	spearman_rho
austin	generous	11	0.991
austin	moderate	11	0.991
austin	scarce	11	0.982
chicago	generous	11	0.955
chicago	moderate	11	0.936
chicago	scarce	11	0.955
nyc	generous	11	0.964
nyc	moderate	11	0.964
nyc	scarce	11	0.964
sf	generous	11	0.882
sf	moderate	11	0.900
sf	scarce	11	0.900

Validation selection by MAE versus by decision loss (frozen budgets):

city	regime	selected_by_val_mae	selected_by_val_decision	test_loss_mae_choic
austin	scarce	global/calendar_weather/lgbm_point	local/calendar_weather/lgbm_point	16572872.00
austin	moderate	global/calendar_weather/lgbm_point	local/calendar_weather/lgbm_point	10869184.00
austin	generous	global/calendar_weather/lgbm_point	local/calendar_weather/lgbm_point	7085533.00
chicago	scarce	local/calendar_weather/lgbm_point	local/calendar_weather/lgbm_point	38101878.00
chicago	moderate	local/calendar_weather/lgbm_point	local/calendar_weather/lgbm_point	21402749.00
chicago	generous	local/calendar_weather/lgbm_point	local/calendar_weather/lgbm_point	6920807.00
nyc	scarce	local/calendar_weather/lgbm_point	local/calendar_weather/lgbm_point	154437576.00
nyc	moderate	local/calendar_weather/lgbm_point	local/calendar_weather/lgbm_point	91571785.00
nyc	generous	local/calendar_weather/lgbm_point	local/calendar_weather/lgbm_point	27875499.00
sf	scarce	global/calendar_weather/lgbm_point	global/calendar_weather/lgbm_point	30255010.00
sf	moderate	global/calendar_weather/lgbm_point	global/calendar_weather/lgbm_point	16972510.00
sf	generous	global/calendar_weather/lgbm_point	global/calendar_weather/lgbm_point	3952067.00

The complete decision grid (including the labeled, non-bounding hindsight myopic diagnostic) is `tab3_decision_all.csv`; block-bootstrap intervals for the pre-named contrasts are `tab8_inference.csv` (decision) and `tab9_forecast_inference.csv` (forecast), each at block lengths 14/28/56.

6 Negative and Conditional Results

The corrected protocol produced several results that resolved against the more optimistic hypothesis; all are retained as findings:

- **Weather adds nothing in Austin:** +0.001 test MAE on top of calendar features (block-bootstrap interval covers zero), against a 9.3% gain in New York.
- **Pooled training hurts under valid censoring:** with stage-censored pooled fits (Amendment C11), pooling raises test MAE significantly in Austin (+4.1%), Chicago (+6.5%), and New York (+1.8%); San Francisco is neutral. Censored zero-shot transfer degrades MAE by +5% to +149%.
- **The exact point-forecast optimizer loses to proportional apportionment** in all 12 city-regime settings under the neutral objective — the identified mechanism being uninformative surplus placement under degenerate forecast distributions.
- **Quantile intervals are under-dispersed:** 90% nominal, 76.8–81.2% empirical; unrepaired in this paper.
- **A protocol defect was detected and corrected during review (R1-F1):** per-city split boundaries differ by 1–2 calendar days, so uncensored pooled fits contained $\approx 0.1\%$ contemporaneous other-city rows. After the targeted censoring fix, the selection-experiment story changed from 7/12 disagreements with 4 apparent harms to 9/12 agreements, 2 gains, 0 harms — the earlier disagreement was substantially an artifact of the contaminated pooled model’s inflated validation scores. The pre-fix outputs are preserved in the repository history for audit.
- **The hindsight myopic reference is not a bound:** being per-day myopic under carryover it can be exceeded; it appears only in the diagnostic table below and is never used as a denominator.

7 Sensitivity Analyses

Full numbers in `outputs/tables/tab7_sensitivity.csv`.

- **Normative-priority scenario** (illustrative, not socially validated): the proportional-optimizer gap compresses relative to the equal-weights primary; the main text therefore states the proportional-dominance claim as conditional on the neutral objective.
- **One-family-at-a-time service yield** ($\times 0.70$ and $\times 1.30$ for every active family, 1,125 sensitivity simulations): no policy-ordering reversal anywhere; the largest perturbation is New York, generous regime, noise yield $\times 1.30$.
- **10% daily abandonment**: policy ordering unchanged in all four cities.
- **Budget calibration on train+validation** instead of training only: conclusions unchanged.
- **Austin without the residual *other* family** (budgets recomputed train-only over the seven retained families): identical policy ordering to the primary analysis in all three regimes; Austin’s conclusions do not depend on its residual category.
- **Bootstrap block length** 14/28/56 days: 0 of 48 decision contrasts and none of the forecast contrasts change conclusion.

8 Computational Resources

All experiments run on a single Linux container (no GPU). Measured wall-clock times for the corrected rerun: 311 acquisition 29–54 minutes per run on hosted CI runners (portal-side query latency); weather acquisition under 2 minutes; panel and feature construction under 1 minute; the forecasting stage (404 evaluations including rolling-origin folds, stage-censored pooled models, the dual-fit quantile model, and censored zero-shot transfer) approximately 13 minutes; the decision stage (324 primary simulations, 1,125 sensitivity simulations, selection experiment, and 144 block-bootstrap intervals) approximately 9 minutes; inference and artifact generation under 2 minutes. Python 3.11; package versions pinned in `reproducibility/pip-freeze.txt`; global seed 20260609; the decision layer is Monte-Carlo-free.

9 Ethics and Responsible-AI Statement

Data. All inputs are public administrative aggregates (daily counts of service requests by category) and public weather observations. No individual-level, household-level, or address-level information is acquired, stored, or processed; the acquisition layer requests server-side aggregation precisely so that record-level attributes (including free-text descriptions and locations) never enter the pipeline.

People inside the data. Each request count summarizes residents’ attempts to obtain public services. Two ethical asymmetries follow. First, reported volume under-represents communities with lower propensity or ability to report (Kontokosta et al., 2017; Kontokosta and Hong, 2021); optimizing service against reported volume can therefore divert resources from quiet need toward vocal demand. This study’s city-level action space cannot reallocate across neighbourhoods, which contains but does not eliminate the concern: the family-level analogue (e.g., systematically deprioritizing housing complaints relative to noise) is measured and reported per policy. Second, the

priority weights are normative inputs; we publish them as configuration, sweep them, and explicitly assign their real-world selection to accountable human decision-makers.

Intended and unintended use. The system evaluated here is a research benchmark. It is not a deployable dispatch tool, has no city involvement or endorsement, and the manuscript states this in every section where simulation results appear. Misuse risk (treating simulated capacity numbers as operational guidance) is mitigated by classifying all operational parameters as sensitivity-only and reporting ranges rather than recommendations.

Human oversight. The intended human-AI division of labour — forecasts and allocation suggestions with uncertainty made visible, final allocation authority retained by planners — is described in the discussion and is a design commitment of the broader research programme, not an evaluated capability of this study.

10 Reproducibility Instructions

One-command rebuild. From the repository root:

```
pip install -r requirements.txt
make test          # core tests + protocol guards G1-G13
make all           # panel -> features -> forecasts -> simulated
                   # decisions -> inference -> artifacts
python3 -m pytest tests -q  # full guard suite (33 tests) must pass
```

The guard suite enforces the frozen protocol: training-only budget calibration invariant to test perturbation; validation-only selection invariant to corrupted test metrics; recomputed censor dates on every transfer row plus a fit-time proof manifest for pooled-model stage censoring (Amendment C11); a single fitted model behind all uncertainty arms; structural absence never encoded as zero; exact budget conservation; forbidden-terminology scans; configuration–output agreement; source-identifier consistency; and provenance hashes binding every artifact to its generating run.

Determinism. A single global seed (20260609) controls every stochastic component (NumPy, scikit-learn, LightGBM); expected-value allocation uses a deterministic 99-point inverse-CDF grid rather than Monte Carlo, so decision results are bit-stable given forecasts.

Data provenance. Raw acquired layers are versioned in `data/raw/` together with JSON manifests recording: source authority, dataset id and page, license note, every query URL issued, retrieval timestamps (UTC), uncompressed and compressed SHA-256 checksums, row counts, and — for 311 sources — the portal’s own dataset metadata (`rowsUpdatedAt`, column inventory) at retrieval time. Re-running `make acquire-311 acquire-weather` re-derives the same layers from the sources (counts for recent periods may drift as portals backfill; the manifests pin what this paper used).

Environment. Python 3.11; exact package versions in `reproducibility/pip-freeze.txt`; no GPU required; total pipeline runtime and per-stage costs in Section 8 of this supplement.

Acquisition environment note. The research sandbox used for this project blocks direct access to the four city portals; the 311 acquisition therefore ran through the repository’s GitHub Actions workflow (`.github/workflows/acquire-311-data.yml`) on GitHub-hosted runners, executing the same committed script and pushing the raw aggregates and manifests back to the branch. The first attempt with full-window queries timed out against the NYC portal and was replaced by per-calendar-year chunking; both attempts are preserved in the repository’s Actions history.

References

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